

Detection of Cardiovascular Disease Using Machine Learning Algorithms and Principal Component Analysis

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Abstract—cardiovascular disease is a family of diseases that are the leading cause of death worldwide. Late diagnosis and treatment contribute to the high number of deaths caused by this family of diseases. To increase accessibility for the general public to get diagnosed and treatment, a cardiovascular disease detection system powered by artificial intelligence (AI) can be developed and used. The healthcare sector has already utilized AI for many purposes and has proven to be as reliable as trained professionals. This study aims to develop an AI model that can detect whether a person has cardiovascular disease through analyzing their clinical data. Both the support vector machine and the decision tree classifier algorithm were trained with the Cleveland heart disease dataset and a version of the dataset that has undergone dimensionality reduction through the principal component analysis (PCA) algorithm, with a 75:25 ratio of training and testing data. The performance of the two algorithms was then evaluated using the confusion matrix which found that the PCA algorithm enhanced the performance of both algorithms and that the SVM model paired with PCA achieved the highest performance of 86.67% accuracy, 90% precision, and 79.41% recall.

Keywords—machine learning, artificial intelligence, cardiovascular disease, decision tree, support vector machine

I. INTRODUCTION

Cardiovascular diseases otherwise known as heart disease are the leading cause of death globally [1] with around 9 million people falling victim to this group of diseases in 2019 alone [2]. With the rapid development in the artificial intelligence sector, AI can play a crucial part in providing early treatment of cardiovascular disease through accurate diagnosis.

Artificial intelligence is already involved in the health and medicine sector with many companies and health professionals utilising AI for various purposes such as radiologist that uses AI to help with a patient's diagnosis based on their screening mammography [3].

This study aims to develop, test, and compare methods for diagnosing cardiovascular disease utilising the support vector machine (SVM) and decision tree classification algorithm (DT), two well-known machine learning method that is famous for its effectiveness classification issues. Each model is going to be trained by a dataset, that has gone through dimensionality reduction using principal component analysis

(PCA), and multiple evaluation techniques will be used to gauge the model's performance in terms of accuracy, specificity, and sensitivity. The study will also examine the big potential of AI models for diagnosing cardiac diseases. The findings of this study could have a significant impact on the creation of a precise and trustworthy instruments for the early detection and prevention of cardiovascular disease, which could prevent deaths and reduce the cost of diagnosis and treatment.

II. RELATED WORKS

Machine learning has become increasingly popular in the medical sector, with many utilizing it for diagnosing purposes [4]. Studies into the use of AI in detecting and diagnosis cardiovascular disease has been done numerous of times with various datasets and machine learning algorithms [5]. Datasets of electrocardiogram (ECG), heart sounds, and the Cleveland cardiovascular disease datasets are commonly used in AI powered cardiovascular disease diagnosis studies.

Kwon et al. [6] developed a model that could detect heart failure with preserved ejection fraction using the convolutional neural network (CNN). Similarly, CNN algorithm and ECG data was used to find common signal and patterns that indicate cardiovascular disease [7]. Chowdhury et al. [8] developed a prototype of a digital stethoscope that can detect cardiovascular disease using a microphone that records a patient's heart sound and an AI model obtaining an accuracy level of 94.63%. Also utilizing heart sound, Sugiyarto et al. [9] could classify cardiovascular disease from phonocardiogram (PCG) data with an 85% accuracy using the continuous wavelet-convolutional neural network algorithm.

The Cleveland cardiovascular disease dataset provided by the UCI machine learning repository, which contains 303 records and 76 features [10], is one of the most popular cardiovascular disease datasets used to train various machine learning algorithms. Chang et al [11] created an optimized random forest algorithm to train on the dataset and achieved 83% accuracy level. Neural networks such as CNN [12], DNN [13], and modified versions of the algorithms [14] has been used on the dataset with each getting accuracy levels higher than 80%. The dataset has also been used numerous times to compare the performance of various algorithms. SVM has been compared with KNN [15], random forest [16],

decision tree classifier [17], gradient boosting [18], and Naïve Bayes [17]. Alalawi et al. [19] compared the performance of 8 different algorithms and found that gradient boosting was the most accurate. Mohan et al. [10] created a hybrid algorithm that combined the random forest algorithm with the linear model and achieved an accuracy level of 88.4%. Haq et al. [20] studied the performance of machine algorithms paired with feature selection on the dataset and found that the combination of relief feature selection along with linear regression had the highest performance.

III. METHODS

In this study we will compare the accuracy of two AI models trained using the SVM algorithm and the decision tree classification algorithm at detecting cardiovascular disease when trained with a dataset containing clinical data, and to achieve this we propose the following methodology which contains dataset, data pre-processing, dimensionality reduction, classification, and evaluation.

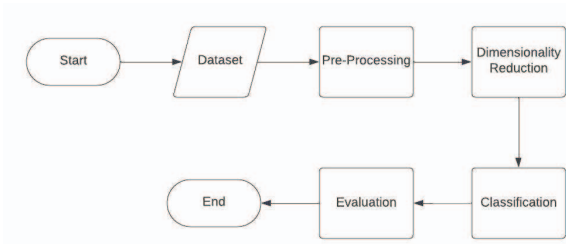


Fig. 1. Methodology flowchart

A. Dataset

The Cleveland cardiovascular disease dataset [21] obtained from Kaggle will be used in this study, as this dataset is popular among similar studies. The dataset contains 76 variables varying from personal information such as age and sex, to clinical data such as resting blood pressure and maximum heart rate achieved, and 1 target parameter that states whether the attributes indicate cardiovascular disease or not, but only 13 feature variable and 1 target variable will be used, as shown in Table 1. The dataset contains a total of 297 records after invalid records have been removed.

TABLE I. DATASET VARIABLES

No.	Dataset Parameters	
	Variable Name	Description
1.	Age	Age in years
2.	Sex	0 = Female 1 = Male
3.	Chest pain type	1 = typical angina 2 = atypical angina 3 = non-anginal pain 4 = asymptomatic
4.	Resting blood pressure	In mmHG
5.	Cholesterol level	In mg/dL
6.	Fasting blood sugar	0 = (< 120 mg/dL) 1 = (> 120 mg/dL)
7.	Resting ECG result	0 = normal 1 = ST-T wave abnormality 2 = probable or definite left ventricular hypertrophy by Estes' criteria
8.	Maximum heart rate	In beats per minute (bpm)

No.	Dataset Parameters	
	Variable Name	Description
9.	Angina due to exercise	0 = No 1 = Yes
10.	ST depression induced by exercise compared to resting condition	0 – 6.2
11.	Slope of ST segment	0 = Upsloping 1 = Flat 2 = Downsloping
12.	Number of major arteries coloured by fluoroscopy	0 - 3
13.	Thallium scan	0 = Normal 1 = Fixed defect 2 = Reversible defect
14.	Condition	0 = No cardiovascular disease 1 = Cardiovascular disease

B. Data Pre-processing

During this step the data will undergo standardization to ensure the classification algorithm can perform optimally. Standardization is the process whereby data is scaled to modify the mean into 0 and have a standard deviation of 1. The dataset will then be split into a 75:25 ratio of training and testing data respectively.

C. Dimensionality Reduction

Principal Component Analysis (PCA) is a popular method of dimensionality reduction, particularly when dealing with large datasets [22]. PCA transforms the original data into a new coordinate system, whereby each axis in that coordinate system is a principal component that captures the maximum amount of variance in the data. Due to the 13 feature variables in the dataset, PCA can help reduce the dimensionality of the data by transforming it into a new coordinate system with fewer variables, while still retaining most of the important information. This can lead to improvements in computational efficiency and model performance, as models trained on the reduced dataset will be faster and more accurate than those trained on the original dataset.

D. Classification

In this study, two classification algorithms will be trained and evaluated with the Cleveland cardiovascular disease dataset. The SVM algorithm and decision tree classification algorithm are two widely known classification algorithms that employ different methods of classification each with their own strengths and weakness.

The support vector machine algorithm is a supervised machine learning algorithm that can perform classification on a labelled dataset. The SVM algorithms has various kernel that are responsible for projecting the data into a higher dimension so that a hyperplane can classify the data in any circumstances [23]. For this study the SVM algorithm will use the linear kernel.

Decision tree classifier is an algorithm that takes a complex decision-making process and breaks it down to simpler processes. It has many practical uses such as speech recognition and medical diagnosis [24]. Decision tree classifier uses entropy to determine the distribution of a given variable with the as shown in equation (1), where H is the entropy and p is the probability of each outcome.

Information gain is then determined by the difference in the entropy value before and after a variable is split, which

can be represented by formula (2). In formula (2) D represents the dataset parent node, $H(D)$ represents the entropy of the dataset, $H(D_v)$ represents the entropy of the dataset split, and D_v is the dataset at a child node upon splitting.

$$H(p) = -\sum_i p_i \log_2 p_i. \quad (1)$$

$$\text{Gain}(D, x) = H(D) - \sum_{v \in \text{Values}(x)} \frac{|D_v|}{|D|} H(D_v). \quad (2)$$

E. Evaluation

The confusion matrix will be used to assess the performance of each classification algorithm. The confusion matrix compares the prediction to the actual values and contain 4 different classes which are the true positive (TP) where input was correctly predicted as positive, false positive (FP) where input was incorrectly predicted as positive, true negative (TN) where input was correctly predicted as negative, and false negative (FN) where input was incorrectly predicted as false [25].

Through the data provided by the confusion matrix we can evaluate the model's accuracy, precision, and recall. The model's accuracy is the ratio of the total correct prediction made by the total prediction made. Precision is the percentage of data that the model correctly predicted as meanwhile recall is the model's true positive divided by its true positive and false negative [26].

IV. RESULT AND DISCUSSION

An analysis on the Cleveland dataset and the performance of the SVM and decision tree classification algorithm has been conducted by following the proposed methodology. Using a correlation matrix it is shown that the thallium scan data has the highest correlation with whether the patient has cardiovascular disease or not, meanwhile the fasting blood sugar data has 0% correlation. Through dimensionality reduction by PCA unimportant data such as the fasting blood sugar can be removed allowing for better performance at classification. To ascertain the effectivity of PCA the method the training was done with and without dimensionality reduction.

A. Support vector machine results

The SVM algorithm, when trained with data that has not gone through dimensionality reduction, correctly predicted 28 patient's data as having cardiovascular disease and correctly predicted a lack of cardiovascular disease in 35 patient's data out of 75 testing data. The model achieved 84% accuracy, 82.35% precision, and 82.35% recall as seen in table 2.

When given the data that has gone through dimensionality reduction, the SVM algorithm correctly predicted 27 patient's data as having cardiovascular disease and 38 patient's data as not having cardiovascular disease out of 75 testing data. Table 2 shows the model achieving 86.67% accuracy, 90% precision, and 79.41% recall.

TABLE II. SVM MODEL RESULTS

Model Type	Accuracy	Precision	Recall
SVM without PCA	84%	82.35%	82.35%
SVM with PCA	86.67%	90%	79.41%

B. Decision tree classifier

The decision tree classifier model achieved 65.33% accuracy, 61.11% precision, and 64.71% recall when trained and tested on data that has not been altered. It correctly predicted 22 patient's data as having cardiovascular disease and 27 patient's data as not having cardiovascular disease out of the 75 patient's data that were used for testing.

After reducing the dimensionality of the data with PCA the model correctly predicted 25 occurrences of cardiovascular disease and 29 occurrences of no cardiovascular disease from 75 patient's data. The model achieved 72% accuracy, 67.57% precision, and 73.53% recall as shown in table 3.

TABLE III. DECISION TREE MODEL RESULTS

Model Type	Accuracy	Precision	Recall
DT without PCA	65.33%	61.11%	64.71%
DT with PCA	72%	67.57%	73.53%

C. Discussion

Through observations of the results, it can be concluded that reducing the dimensionality by using the PCA algorithm has benefited the performance of both classifiers. Table 4 shows SVM model achieved an increase of 2.67% accuracy meanwhile the decision tree classifier achieved an increase of 6.67% in accuracy. By reducing the dimensionality, the model can be prevented from being overfitted which can cause it to not perform as well when given new data.

From Table 4 it can also be observed that the SVM outperformed the decision tree classifier in all aspects with and without PCA being used on the dataset. Therefore, for the development of cardiovascular disease detection systems that implement AI it would be better to utilize the SVM algorithm combined with the PCA algorithm compared to the decision tree classifier, but it cannot be concluded from this study that the SVM algorithm would be best suited for such a system. Due to limitations present in the studies.

TABLE IV. SUMMARY OF MODEL PERFORMANCE

Model Type	Accuracy
SVM without PCA	84%
SVM with PCA	86.67
DT without PCA	65.33%
DT with PCA	72%

V. CONCLUSION

This study was conducted to determine whether the support vector machine and decision tree classifier can detect cardiovascular disease from a dataset of clinical data and to compare the performance of each algorithm when paired with PCA as a method of dimensionality reduction. To do this, the Cleveland cardiovascular disease dataset was standardized, split into training and testing data, and underwent dimensionality reduction by means of the PCA algorithm. The two classification algorithms were then trained on the data and tested using the confusion matrix.

When the proposed methodology was conducted it was found that the support vector machine algorithm with PCA performed the best, with an accuracy of 86.67%. Meanwhile

the decision tree classifier achieved an accuracy of 72% when paired with PCA. Although the SVM algorithm outperformed the decision tree classifier it cannot be said that the algorithm can be utilized for a cardiovascular disease detection system as considering the responsibilities risk of such a system, 86.67% is not accurate enough to be relied upon.

Limitations within this study such as the comparison of only two classification algorithm, lack of testing using different dimensionality reduction algorithm such as the linear discriminant analysis, and the small quantity of data in the dataset used are hoped to be addressed by future studies. The authors of this study hope that through this article, advancement in the cardiovascular disease treatment sector can be made to prevent such a prevalent cause of death.

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